

# Allan Vikiru

## Machine Learning | Edge Computing | Internet of Things

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### Skills and Competencies

- Software Engineering
- Machine Learning
- Hardware Prototyping
- Spatial Data Analysis
- IoT Prototyping
- Data Engineering
- Web and Cloud Development
- Object-Oriented Programming
- Commercial and Academic Presentations
- Technical Writing
- Project Management
- Stakeholder Engagement

Languages: C++, JavaScript, LATEX, MATLAB, Python, R

Software: Arduino, InfluxDB, Git, Django, Heroku, Microsoft Power BI, QGIS, Streamlit

### Education

#### **MSc. Information Technology (Business Intelligence & Data Analytics Option) [2024]**

Strathmore University - Nairobi, Kenya

Thesis Title: A Predictive Model for Determining Vegetable-Derived Macronutrients for a Diabetic Patient

#### **BSc. Informatics & Computer Science [2021]**

Strathmore University - Nairobi, Kenya

Final Year Project Title: Determining the Risk of Fireground Injuries among Firefighters using IoT

### Work Experience

#### **Tutorial Fellow – School of Computing and Engineering Sciences, Strathmore University [July 2024 – Present]**

- Teaching at least 250 students on Artificial Intelligence, Embedded Systems & IoT, Information Security, Data Governance and Cloud Computing.
- Supervising up to 25 students on projects related to challenge-driven education, deep learning and Internet of Things.

#### **Research Coordinator – SCES Makerspace Lab, Strathmore University [July 2023 – Present]**

- Continually engaging a team of 6 faculty researchers and at least 10 undergraduate and graduate students in developing computing and engineering projects.
- Co-lead of the **Conservation Technology Research Hub**. Key tasks include internal and external stakeholder engagement, data collection and analysis, prototyping, grant proposal writing and conducting team meetings.

#### **Graduate Assistant – School of Computing & Engineering Sciences, Strathmore University [March 2022 – June 2024]**

- Developed a research proposal on using machine learning and IoT to detect logging activities that received an award of USD 139,500.
- Contributor to three research papers on deep learning and cloud computing.

#### **Software Developer Intern – IoT & Wireless Networks Lab, @iLabAfrica [January 2022 - March 2022]**

- Designed and implemented a monitoring dashboard in Vue.js and Node.js and an authentication system in Keycloak for a LoRaWAN-based research project.
- Designed and developed breadboard model devices to model LoRaWAN connectivity.

## Publications

**Vikiru A.** & Khakata E. (in press). A Wearable Sensor System for Monitoring the Risk of Fireground Injuries among Firefighters.

11th International Congress on Information and Communication Technology.

Simiyu, D., Muchiri, H., Vikiru, A., Butime, J., & Muti, Z. (2024). A Forest Acoustics – Temporal Frequency Convolution Neural Network Model for Detecting Illegal Logging Activities in Forest. 2024 5th International Conference on Smart Sensors and Application (ICSSA), 1–6. <https://doi.org/10.1109/ICSSA62312.2024.10788639>

Ayankoso, S., Wang, Z., Shi, D., Yang, W., **Vikiru, A.**, Kamau, S., Muchiri, H., & Gu, F. (2024). Development of Long-Range, Low-Powered and Smart IoT Device for Detecting Illegal Logging in Forests. *Journal of Dynamics, Monitoring and Diagnostics*. <https://doi.org/10.37965/jdmd.2024.550>

Simiyu, D., **Vikiru, A.**, Muchiri, H., Gu, F., & Butime, J. (2024). A Chainsaw-Sound Recognition Model for Detecting Illegal Logging Activities in Forests. In A. D. Ball, H. Ouyang, J. K. Sinha, & Z. Wang (Eds.), *Proceedings of the UNified Conference of DAMAS, IncoME and TEPEN Conferences (UNified 2023)* (pp. 797–806). Springer Nature Switzerland. [https://doi.org/10.1007/978-3-031-49421-5\\_65](https://doi.org/10.1007/978-3-031-49421-5_65)

**Vikiru, A.** (2023, March 18). *Market Basket Analysis using Apriori Algorithm*. <https://allanvikiru.github.io/MarketBasketAnalysis/>

**Vikiru, A.**, Muiruri, M., & Ateya, I. (2023). *An Overview on Cloud Distributed Databases for Business Environments* (arXiv:2301.10673). arXiv. <https://doi.org/10.48550/arXiv.2301.10673>

## Conferences & Workshops

### **Delegate – GDH Partner Days on Challenge Driven Education**

Royal Institute of Technology (KTH) - Stockholm, Sweden

March 2026

### **Presenter - A Wearable Sensor System for Monitoring the Risk of Fireground Injuries among Firefighters**

11th International Congress on Information and Communication Technology - London, United Kingdom

February 2026

### **Delegate – Microsoft Faculty Skilling and Immersion Program (Cohort 3 – AI-Oriented Education)**

Microsoft ADC & Microsoft Leap - Nairobi, Kenya

June 2025

### **Delegate - 5th International Conference on Smart Sensors and Application**

Universiti Teknologi Malaysia (UTM) - Penang, Malaysia

September 2024

### **Project Administrator - Curbing Illegal Logging Patterns Using Sound-Based Detection Techniques**

Strathmore University (Kenya), Université de Neuchâtel (Switzerland), University of Huddersfield (United Kingdom), Kenya Forestry Research Institute (Kenya)

July 2023 - September 2024

### **Delegate – CMU Innovation Summit 2024**

Carnegie Mellon University Africa - Kigali, Rwanda

January 2024

### **Organiser & Presenter – Fundamentals of Raspberry Pi 400**

Strathmore University - Nairobi, Kenya

November 2023